

Functions and series

Foundations for sets and mappings

Key concepts

Terminology

Types of numbers

Domains

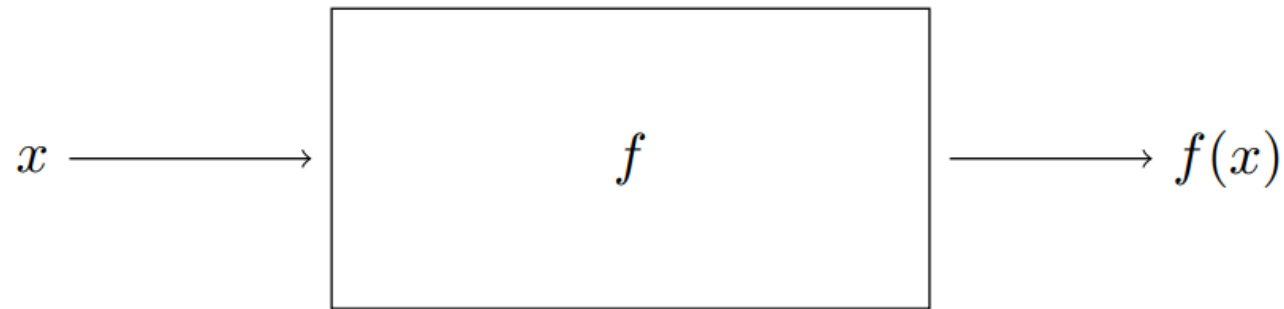
Images

Arithmetic progression

Geometric progression

Function notation

A function from a set X to a set Y is a rule that associates to each element x from X a unique element y from Y . If we denote our function by f , then the unique point y associated to x is usually written $f(x)$, or $y = f(x)$.



Note that the letter x is simply a symbol that represents a value that is passed into the function f . We could also have used any symbol instead. Other letters (symbols) that are frequently used for this task are t , y , and s .

Example Functions

A function is one of the most important concepts in Maths

You already know loads e.g.

The doubling function: $x \rightarrow 2x$, or $y = 2x$, or $f(x) = 2x$

The squaring function: $x \rightarrow x^2$, or $y = x^2$, or $f(x) = x^2$

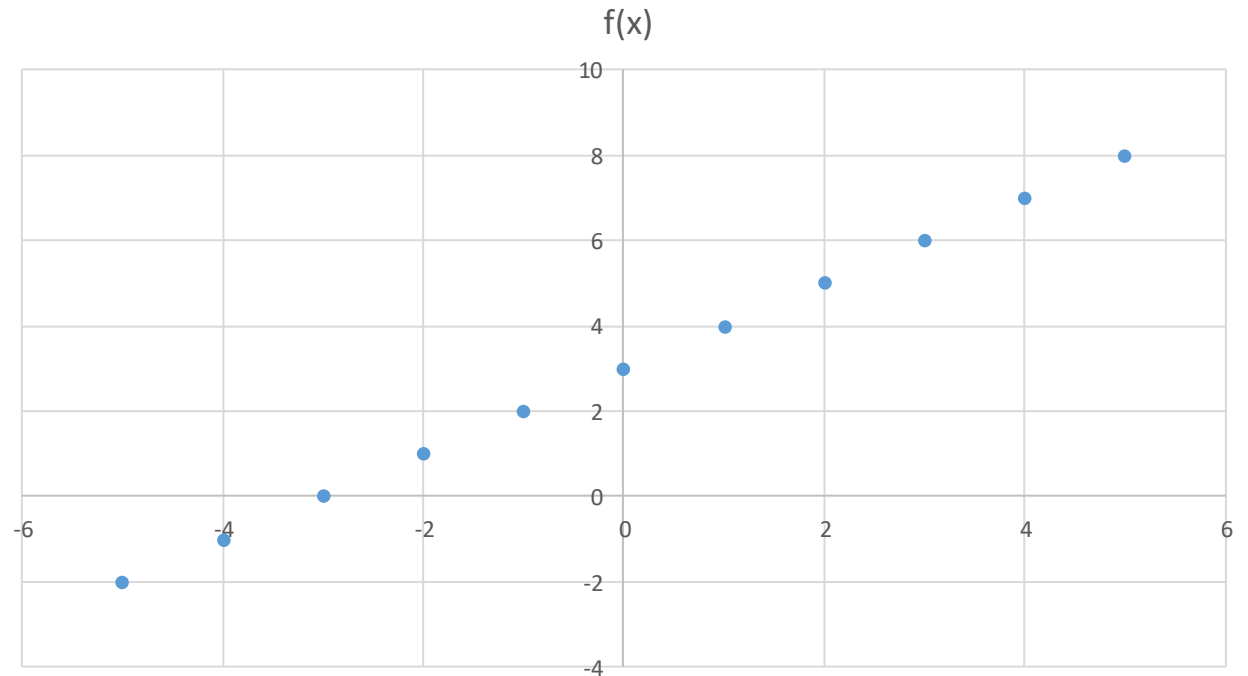
The sine function: $x \rightarrow \sin x$, or $y = \sin x$, or $f(x) = \sin x$

Graph of a function

We can also represent the function using its graph, which is the set of points $(x, f(x))$ on the x - y -axis.

e.g. $f(x) = x + 3$

What about the domain
of the function?



Domains

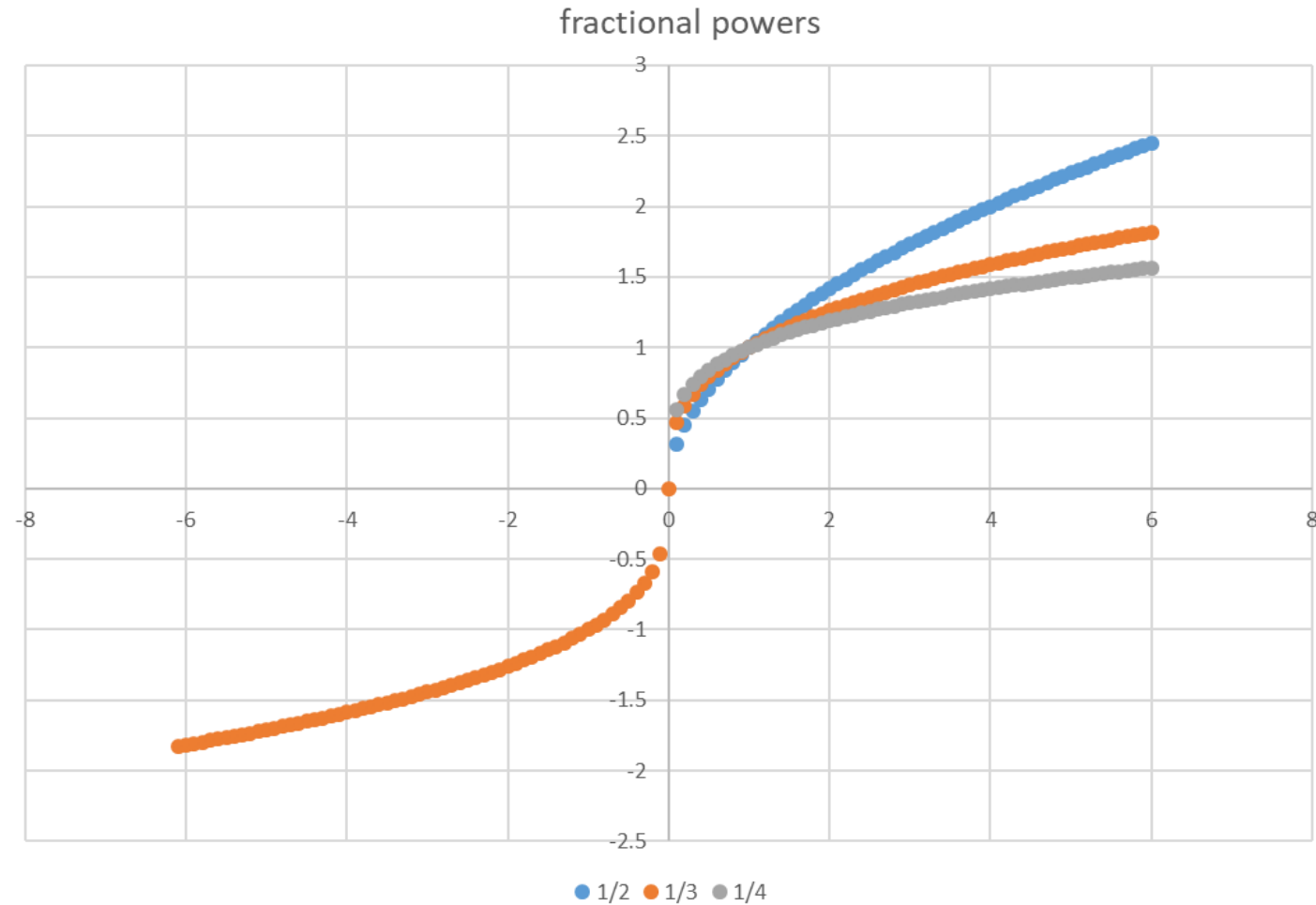
Domain is the range of input values a function can have.

Some functions are only valid over particular domains

Last week we looked at the fraction powers functions:

We simply can't find $\sqrt[2]{-2}$

However, we sometimes omit the domain and instead assume that the function is defined wherever that formula is valid—usually this means that the formula does not involve a division by zero.



Types of numbers

Natural Numbers

$\mathbb{N}$

The numbers children are fine with: 1, 2, 3... etc. (may include zero)

Integer Numbers

$\mathbb{Z}$

As natural, but including negative numbers: ..., -2, -1, 0, 1, 2...

Rational Numbers

$\mathbb{Q}$

Any number that be represented as a ratio of two integers: $\frac{1}{2}$, $\frac{6}{1}$, $\frac{3}{7}$...

Real Numbers

$\mathbb{R}$

Any number that could be used as a distance on a line (so excludes complex numbers)

Irrational Numbers

$\mathbb{P}$

Not a well defined set, sometimes written as $\mathbb{R}-\mathbb{Q}$ includes π , e , $\sqrt{2}$ etc. Numbers with un-repeating, endless decimals

Sets

The types of numbers above indicate different sets.

Notation uses set brackets (braces): $\{1,2,3,4\}$

This is the set of positive integers between zero and five, or between 1 and four, inclusive, so we write:

$$\{x | 1 \leq x \leq 4\}$$

The empty set has no elements e.g. the number of fish that drive BMWs

It is denoted $\{\}$

The empty set is a subset of every set, including itself

Mapping

The **doubling** function maps from $\mathbb{R}$ to $\mathbb{R}$

The **squaring** function maps from $\mathbb{R}$ to the set of non negative numbers $\mathbb{Q}^+$

The **sine** function maps from $\mathbb{R}$ to the interval $[-1,1]$

We write this $\{x \mid -1 \leq x \leq 1\}$

Image

The image of a function is the set of all values that can be obtained using the function; the set of all $f(x)$ as x ranges over all valid input values.

The *sin* and *cos* functions are different and give distinct values for any given x value but their image is the same: $\{x \mid -1 \leq x \leq 1\}$

Piecewise functions

Some functions cannot be defined using a single formula. Instead, we must split the domain into distinct parts where we can use a formula. These functions can be defined piecewise (piece-by-piece). For example, the real function f defined by:

$$f(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0 \end{cases}$$

This function assigns the value 1 to all positive numbers, the value -1 to all negative numbers, and the value 0 to 0, it cannot be defined everywhere by a single formula

Series

Series are lists of functions that follow some kind of pattern. Consider this sequence:

$$a + (a + d) + (a + 2d) + (a + 3d) + \dots$$

We can number the terms, from 1 to n :

$$\begin{array}{ccccccc} a & + & (a + d) & + & (a + 2d) & + & (a + 3d) + \dots \\ n = 1 & & n = 2 & & n = 3 & & n = 4 \quad + \dots \end{array}$$

The last term is given by: $+(a + (n - 1)d)$.

Arithmetic progression

Where consecutive terms of a sequence differ by a constant value, such a sequence is referred to as an **Arithmetic Progression** (A.P).

We define:

a = 1st term in the sequence

l = last term in the sequence

d = common difference between each term

n = number of terms in the sequence

Averaging

$$x + (x + 4) + (x + 2 \times 4) + \cdots + (x + (n - 1) \times 4)$$

The difference between each term is constant (in this case, 4) so $d = 4$

We can find the average value of a term by averaging the first and last terms.

The difference between terms (d) is 4

The first term of the sequence (a) is x

The last term of the sequence (l) is $(x + (n - 1) \times 4)$

So the average term is $\frac{2x + 4(n - 1)}{2} = x + 2(n - 1)$

Summation

We can multiply the average term by the number of terms to give the sum of an Arithmetic progression:

$$S_n = \left(\frac{a + l}{2} \right) n$$

a = 1st term in the sequence

l = last term in the sequence

d = common difference between each term

n = number of terms in the sequence

Example summation 1

What is the sum of the following sequence: 2 + 6 + 10 +14?

$$S_n = \left(\frac{a + l}{2} \right) n = \left(\frac{2 + 14}{2} \right) \times 4 = 32$$

Yes, I know it was an easy one, but what works for four numbers works for thousands. It also lets us start to think about the series parameters...

Example summation 2

What about: $2+6+\dots+50$?

$$S_n = \left(\frac{a + l}{2} \right) n = \left(\frac{2 + 50}{2} \right) \times n$$

We can define n as $n = \left(\frac{l-a}{d} \right) + 1$

So

$$S_n = \left(\frac{a+l}{2} \right) \left(\frac{l-a}{d} \right) + 1 = \left(\frac{2+50}{2} \right) \times \left(\frac{50-2}{4} + 1 \right) = 338$$

Known n

If we are given only a fraction of the sequence, which includes the first term and the common difference, and we are told the number of terms, we can still calculate the sum of the sequence without the last term:

$$\frac{n}{2}(2a + (n - 1)d)$$

e.g. $2+6+10+\dots, n=8$

$$\frac{n}{2}(2a + (n - 1)d) = \frac{8}{2}(2 \times 2 + (8 - 1) \times 4) = 128$$

Unknown d

Usually, we can spot d **by inspection** but we can also calculate it if we know the other quantities:

$$d = \left(\frac{l - a}{n - 1} \right)$$

Other than by inspection, show that the common difference of the following sequence is, in fact, 4.

$2 + 6 + 10 + \dots + 30$, $n = 8$. From this sequence we can state $a = 2$, $l = 30$ and $n = 8$ (which is given).

$$d = \left(\frac{l - a}{n - 1} \right) = \left(\frac{30 - 2}{8 - 1} \right) = 4$$

Summation notation

Writing out long lists of additions is tedious, so we'll use the summation notation, sigma, that we met looking at the standard deviation.

$$3 + 6 + 9 + 12 + 15 + 18 + 21 + 24 + 27 + 30 + 33 + 36 =$$

$$\sum_{i=0}^{n-1} a + d_i = \sum_{i=0}^{n-1} 3 + 3_i, \quad \text{or} \quad \sum_{i=1}^n 3_i$$

Geometric progression

Arithmetic progressions have a common difference but geometric progressions have a common ratio:

$$a + ar + ar^2 + ar^3 + ar^4 + \dots + ar^{n-1} = \sum_{i=1}^n a \left(\frac{1 - r^n}{1 - r} \right)$$

Think about what happens if $r = 1$

Think about a special case of this, where $r < 1$

In fact, think about the **modulus**, $|r| < 1$

In this case, the sum of an infinite series is given by:

$$S_{\infty} = \sum_{i=1}^{\infty} a \left(\frac{1}{r - 1} \right) = \sum_{i=1}^{\infty} \frac{a}{r - 1}$$

Geometric progression example

Find the sum of the series: $3 + 6 + 12 + \dots + 3072$

We can put this into the form for a geometric progression:

$$a + ar + ar^2 + ar^3 + ar^4 + \dots + ar^{n-1}$$

$$3 + (3 \times 2) + 3 \times 2^2 + 3 \times 2^3 + \dots + 3 \times 2^{10} *$$

* Struggle to find that 3072 is 3×2^{11} ? Look back at the exponential work from last Friday!

Geometric example continued

$$3 + (3 \times 2) + 3 \times 2^2 + 3 \times 2^3 + \dots + 3 \times 2^{10}$$

Since $n-1 = 10$, we know that $n = 11$

$$\sum_{i=1}^n a \left(\frac{1 - r^n}{1 - r} \right) = \sum_{i=1}^{11} 3 \left(\frac{1 - 2^{11}}{1 - 2} \right) = 6141$$

Learning process

This is not an easy or intuitive topic. It is an important one, because it lets us describe long sequences in just a few characters and that allows us to be more computationally efficient when coding.

There is more help and additional examples in Chapter 12, page 126 (and chapter 13, p141) of Foundation Maths (Tony Croft)